

* Fig 3.17 (g)

- Shows that requiring $T(r)$ to be **strictly monotonic** ensures the inverse mappings will be **single valued**; means the mappings will be one-to-one in both directions)
- Since images are stored using intensity values
 - Must round values to the nearest integer
 - Doing so results in **strict monotonicity not being satisfied** and implies that the inverse transformation may not be unique
 - The good news: this isn't difficult to handle as shown in example 3.7

* The intensity of an image

- Can be viewed as a random variable within the interval of $[0, L-1]$
- Let $p_r(r)$ and $p_s(s)$ denote the **Probability Density Function (PDF)** of intensity values r and s in two different images
 - The subscripts on p indicate two different functions
 - This result from the probability theory that if $p_r(r)$ and $T(r)$ is known and $T(r)$ is continuous and differentiable over the range of values of interests
 - The PDF value s can be calculated as: $p_s(s) = p_r(r) \left| \frac{dr}{ds} \right|$
 - In short: The PDF of the **output intensity s** is determined by:
 - 1) The PDF of the **input intensities**
 - 2) The **transformation functions used**

What Does PDF Represent in Image Processing?

In image processing—especially in the context of histogram processing and equalization—you can think of a PDF as a way to describe how intensity values are distributed in an image.

More formally:

- The PDF of an image's intensity values gives you the probability that a randomly selected pixel has a certain intensity value.
- If an image is 8-bit, it has intensity values ranging from 0 to 255.
- The PDF $p(r)$ tells you how likely each of those intensity values is.

EXAMPLE 3.7: Illustration of the mechanics of histogram specification.

Consider the 64×64 hypothetical image from Example 3.5, whose histogram is repeated in Fig. 3.22(a). It is desired to transform this histogram so that it will have the values specified in the second column of Table 3.2. Figure 3.22(b) shows this histogram.

The first step is to obtain the histogram-equalized values, which we did in Example 3.5:

$$s_0 = 1; s_1 = 3; s_2 = 5; s_3 = 6; s_4 = 6; s_5 = 7; s_6 = 7; s_7 = 7$$

In the next step, we compute the values of $G(z_k)$ using the values of $p_r(z_k)$ from Table 3.2 in Eq. (3-21):

$$\begin{aligned} G(z_0) &= 0.00 & G(z_1) &= 0.00 & G(z_2) &= 2.45 & G(z_3) &= 5.95 \\ G(z_4) &= 0.00 & G(z_5) &= 1.05 & G(z_6) &= 4.55 & G(z_7) &= 7.00 \end{aligned}$$

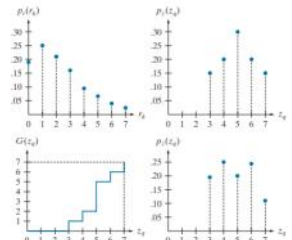
As in Example 3.5, these fractional values are rounded to integers in the range $[0, 7]$:

$$\begin{aligned} G(z_0) &= 0.00 \rightarrow 0 & G(z_1) &= 2.45 \rightarrow 2 \\ G(z_2) &= 0.00 \rightarrow 0 & G(z_3) &= 4.55 \rightarrow 5 \\ G(z_4) &= 0.00 \rightarrow 0 & G(z_5) &= 5.95 \rightarrow 6 \\ G(z_6) &= 1.05 \rightarrow 1 & G(z_7) &= 7.00 \rightarrow 7 \end{aligned}$$

These results are summarized in Table 3.3. The transformation function, $G(z_k)$, is sketched in Fig. 3.22(c). Because its first three values are equal, G is not strictly monotonic, so condition (a') is violated. Therefore, we use the approach outlined in Step 3 of the algorithm to handle this situation. According to this step, we find the smallest value of z_k so that the value $G(z_k)$ is the closest to s_k . We do this for every value of

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FIGURE 3.22 (a) Histogram of a 3-bit image. (b) Specified histogram. (c) Transformation function obtained from the specified histogram. (d) Result of histogram specification. Compare the histograms in (b) and (d).



* One transformation function that's important in Image Processing

Remember: r is the intensity of an image that's going to be processed

- So $T(r) = (L-1) \int_0^r p_r(w) dw$ w is dummy variable of integration
- The integral on the right: **Cumulative Distribution Function (CDF)** of the random value r
- Since PDFs always are positive
 - And the integral of a function is the area under the curve
 - These two conditions satisfy Equation 3-11 since
 - 1) The area under the function **can't decrease as r increases**
 - When the upper limit of this equation is **$r = (L-1)$** , the integral evaluates to 1, as it must for a PDF
 - So both the max value of s is $L-1$ and the stuff above the upper limit is satisfied

$$s = T(r) = (L-1) \int_0^r p_r(w) dw \quad (3-11)$$

* Using Eq. 3-10

- To find $p_s(s)$
- Taking the **Leibniz's rule** and substituting the result for $\frac{dr}{ds}$ in Eq. 3-10 gives the following

$$\begin{aligned} \frac{ds}{dr} &= \frac{dT(r)}{dr} \\ &= (L-1) \frac{d}{dr} \left[\int_0^r p_r(w) dw \right] \\ &= (L-1) p_r(r) \end{aligned} \quad (3-12)$$

$$\begin{aligned} p_s(s) &= p_r(r) \left| \frac{dr}{ds} \right| \\ &= p_r(r) \left| \frac{1}{(L-1) p_r(r)} \right| \\ &= \frac{1}{L-1} \quad 0 \leq s \leq L-1 \end{aligned} \quad (3-13)$$

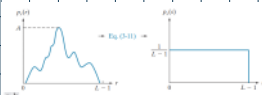


FIGURE 3.18 (a) An arbitrary PDF. (b) Result of applying Eq. (3-11) to the input PDF. The resulting PDF is always uniform, independent of the shape of the input.

$$p_s(s) = p_r(r) \left| \frac{dr}{ds} \right| \quad (3-10)$$

We recognize the form of $p_s(s)$ in the last line of this equation as a **uniform probability density function**. Thus, performing the intensity transformation in Eq. (3-11) yields a random variable, s , characterized by a uniform PDF. What is important is that $p_s(s)$ in Eq. (3-13) will **always** be uniform, **independently** of the form of $p_r(r)$. Figure 3.18 and the following example illustrate these concepts.

Key Takeaways from the Section

1. Intensity Values Can Be Treated Like Random Variables

- Why it matters: This allows us to use concepts from probability theory (like PDFs and CDFs) to analyze and manipulate images mathematically.
- Practical meaning: You can describe how brightness values (intensities) are distributed using a Probability Density Function (PDF), and then transform them to enhance the image.

2. Intensity Transformations Are Defined as Functions:

$$s = T(r)$$

- r = input intensity
- s = output intensity
- $T(r)$ = transformation function applied pixel-wise

3. Monotonicity is Crucial

- Monotonic increasing ensures that the relative brightness order of pixels is preserved.
- Strict monotonicity is needed for a unique inverse transformation $T^{-1}(s)$.

Elaborate more on what the PDF, CDF

Absolutely! Let's break down PDF and CDF in the context of image processing, especially when applied to histogram equalization:

PDF (Probability Density Function)

Definition:

The PDF tells you the probability of each intensity level appearing in the image.

Conceptually:

- If an 8-bit grayscale image has 256 intensity levels (0 to 255), the PDF tells you **how common** each intensity is.
- It's a **normalized histogram**:

$$PDF[k] = \frac{\text{Number of pixels with intensity } k}{\text{Total number of pixels}}$$

Example:

If 100 out of 10,000 pixels are pure black (intensity 0), then:

$$PDF[0] = \frac{100}{10000} = 0.01$$

3. Monotonicity is Crucial

- Monotonic increasing ensures that the relative brightness order of pixels is preserved.
- Strict monotonicity is needed for a unique inverse transformation $T^{-1}(s)$.
- Why it's important: if your transformation isn't monotonic, your image may get weird intensity inversions (artifacts), or you won't be able to reverse the transformation properly.

5. Histogram Equalization Uses the CDF as the Transformation

- The transformation:

$$s = T(r) = (L - 1) \int_0^r p_r(w) dw$$

is the CDF scaled to the range $[0, L - 1]$.

- Why this matters: This transformation spreads out the histogram values, turning non-uniform histograms into uniform ones.
- Outcome: Enhanced contrast and detail in the image.

6. Performing Histogram Equalization Produces a Uniform PDF

- Regardless of the original histogram (dark, bright, low/high contrast), this transformation yields:

$$p_s(s) = \frac{1}{L - 1}$$

A uniform histogram across all intensity levels.

- Why it's powerful: It stretches out frequently occurring values and compresses less frequent ones, enhancing contrast across the image.

Real-World Use Case: Histogram Equalization

- You'd apply this when:
 - The image is too dark or too bright overall.
 - You want to enhance local detail (like in medical images or satellite photos).
- Example tools: OpenCV's `cvtColor`, or manually compute the CDF and map pixels.

Summary:

Concept	Meaning in Practice
PDF	Histogram (as a probability distribution)
CDF	Accumulated brightness info, used for remapping
Monotonic Transform	Preserves pixel order and avoids artifacts
Histogram Equalization	Enhances contrast by flattening the histogram
Transformation $T(r)$	Maps input intensity to new values via probability theory

Example:

If 100 out of 10,000 pixels are pure black (intensity 0), then:

$$PDF[0] = \frac{100}{10000} = 0.01$$

So:

- The sum of all PDF values across 0 to 255 is always 1.0.
- Think of it as distribution of light across the image.

CDF (Cumulative Distribution Function)

Definition:

The CDF tells you the cumulative probability up to a certain intensity level.

Conceptually:

- It answers: "What percentage of pixels have intensities less than or equal to k ?"

$$CDF[k] = \sum_{i=0}^k PDF[i]$$

- It's **monotonic**: it either stays the same or increases with each step.
- Used in histogram equalization to **redistribute pixel intensities uniformly** across the available range (0 to 255).

Analogy:

Term	Analogy
PDF	Bar graph showing the count/probability of each brightness level
CDF	Cumulative percentage — like how much of the bar chart is "filled in" as you move from dark to light

Example:

Imagine a simplified grayscale image (4 levels: 0 to 3):

Intensity	# of Pixels	PDF	CDF
0	2	0.2	0.2
1	3	0.3	0.5
2	4	0.4	0.9
3	1	0.1	1.0

- PDF shows how common each level is.
- CDF shows how many pixels are $\leq$ that level.

In Histogram Equalization:

We remap pixel values based on the CDF:

$$\text{New Intensity} = \text{round}(CDF[k] \times (L - 1))$$

This spreads the intensity values across the full 0-255 range, enhancing contrast, especially in low-contrast images.

Histogram Equalization Problem

Example Problem

We'll use a grayscale image with 8 pixels and intensity values ranging from 0 to 7. (This simulates a small 3-bit image with intensities in the range [0, 7].)

Input image (1D for simplicity):

Intensity	Copy	Edit
Image pixels: [3, 3, 4, 5, 6, 6, 7, 7]		

Step 1: Compute Histogram

We count how many times each intensity value appears:

Intensity (r)	Frequency
0	0
1	0
2	0
3	2
4	1
5	1
6	2
7	2

Probability of occurrence of intensity level R_k

N_k = Number of pixels with intensity R_k



Step 2: Compute PDF (Probability Density Function)

We normalize the histogram by dividing each frequency by the total number of pixels ($N = 8$):

$$PDF(r) = \frac{hist(r)}{N}$$

r	hist(r)	PDF(r)
0	0	0
1	0	0
2	0	0
3	2	$2/8 = 0.25$
4	1	$1/8 = 0.125$
5	1	$1/8 = 0.125$
6	2	$2/8 = 0.25$
7	2	$2/8 = 0.25$

Step 3: Compute CDF (Cumulative Distribution Function)

We calculate the cumulative sum of the PDF:

$$CDF(r) = \sum_{k=0}^r PDF(k)$$

r	PDF(r)	CDF(r)
0	0	0
1	0	0
2	0	0
3	0.25	0.25
4	0.125	0.375
5	0.125	0.5
6	0.25	0.75
7	0.25	1.0

CDF = Summation

Step 4: Apply the Histogram Equalization Formula

The equalization function is:

$$s = T(r) = \text{round}((L - 1) \cdot CDF(r))$$

Where $L = 8$ (since values go from 0 to 7)

So $s = \text{round}(7 \cdot CDF(r))$

r	CDF(r)	$T(r) = \text{round}(7 \cdot CDF)$
0	0	0
1	0	0
2	0	0
3	0.25	2
4	0.375	3
5	0.5	4
6	0.75	5
7	1.0	7

Step 5: Map Old Intensities to New Intensities

We replace every pixel with its transformed value:

Original pixels:

0sharp Copy Edit
[1, 3, 4, 5, 5, 6, 7, 7]

Mapped values using $T(r)$:

0sharp Copy Edit
[2, 2, 3, 4, 5, 5, 7, 7]

Summary

Step	Purpose
Histogram	Count how often each intensity appears
PDF	Normalize histogram to represent probabilities
CDF	Accumulate those probabilities
$T(r)$	Redistribute intensities to stretch contrast
Output	Equalized image has more balanced contrast

Check out WSL VS Code ImageProcessingBook HistogramEqualization directory for an example

Example 3.6: Histogram Equalization

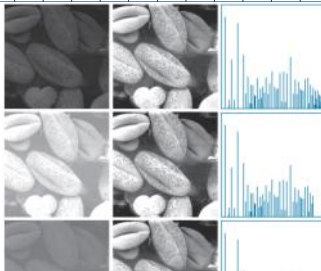
- Left column of Fig 3.20
 - Shows four images from Fig 3.16
 - Center column showcases results from performing histogram equalization on each of the images
 - H.E didn't have much effect on the fourth image since its intensity spans almost the full scale already

Fig 3.21

- Shows transformation function used to generate equalized images using Equation 3-15
- The fourth transformation that's plotted is nearly linear
- Which indicates the inputs mapped nearly equal the outputs

Third column in Fig 3.20

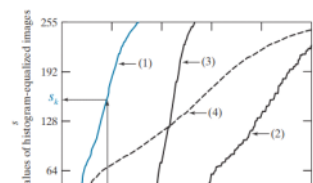
- Shows the histograms of the equalized images
- Despite the histograms being different
 - The histogram equalized images visually look very similar
 - Which is not totally unexpected since the basic difference between the images on the left column is just the contrast rather than the content/pixel values
 - Since the images have the same content
 - Increasing the contrast due to the histogram equalization was enough to render any intensity differences between the equalized images made them look visually



$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k p_r(r_j) \quad k = 0, 1, 2, \dots, L - 1 \quad (3-15)$$

FIGURE 3.21

Transformation functions for histogram equalization. Transformations (1) through (4) were obtained using Eq. (3-15) and the histograms of the images on the left column of Fig. 3.20. Mapping of one intensity value r in



- Despite the histograms being different
 - o The histogram equalized images visually look very similar
 - o Which is not totally unexpected since the basic difference between the images on the left column is just the contrast rather than the content/pixel values
 - o Since the images have the same content
 - Increasing the contrast due to the histogram equalization was enough to render any intensity differences between the equalized images made them look visually indistinguishable
- Judging by the contrast differences between the original images and the equalized ones illustrates the power of histogram equalization as an autonomous contrast enhancement tool

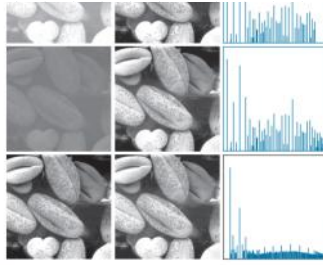


FIGURE 3.20 Left column: Images from Fig. 3.16. Center column: Corresponding histogram-equalized images. Right column: Histograms of the images in the center column (compare with the histogram in Fig. 3.16).

obtained using Eq. (3-15) and the histograms of the images on the left column of Fig. 3.20. Mapping of one intensity value r_i in image 1 to its corresponding value s_i is shown.

